

Daksh Manu Arya

📞 +91 9416129284 ✉ dakshmanuarya@gmail.com 💻 daksh-manu-arya 🌐 GitHub 📍 Noida, U.P.

Professional Summary

Software Development Engineer with experience building distributed systems, Kubernetes-native infrastructure and observability platforms at E2E Networks. Built production backend systems processing over **1M events/min** using Python, Rust, Kubernetes, ClickHouse, OpenTelemetry, and gRPC with a focus on performance, scalability, and reliability.

Technical Skills

Languages: Python, Rust, C/C++, Bash

Backend: Django, Django REST Framework, Celery, gRPC

Cloud & DevOps: Kubernetes, Docker, Helm, Git, GitLab, Temporal

Databases & Messaging: MySQL, ClickHouse, RabbitMQ, NATS JetStream

Observability: OpenTelemetry, OTEL Collector, Grafana, Tempo, Vector

Core CS: Distributed Systems, Operating Systems, DBMS, Computer Networks, DSA, OOP

Experience

E2E Networks Ltd.

Nov 2025 – Present

Software Development Engineer - 1

New Delhi, India

- Developing a **VM autoscaling framework** for **KubeVirt**-based virtual machines running on Kubernetes, designing an **Auto Scaling Group (ASG)** with configurable minimum/maximum replicas, cooldown periods, and CPU, memory, and custom metric-based scaling policies to provision and terminate VM instances automatically.
- Implementing a **VM image management** workflow to capture running virtual machines as reusable images, enabling rapid provisioning and reproducible infrastructure deployments.
- Built a production-grade **Kubernetes observability platform** processing over **1M events/min** through a centralized custom **OTLP gateway** and multi-tenant OpenTelemetry Collector supporting secure telemetry ingestion.
- Reduced authentication latency using a **1M-entry token cache**, achieving **p99 <5ms** cache-hit latency and **p99 <20ms** cache-miss latency while enforcing fail-closed authorization through a Rust gRPC authentication service.
- Engineered a **Rust gRPC** log ingestion and query service backed by **ClickHouse**, integrating **NATS JetStream** and **Vector** for reliable log delivery across bare metal, virtual machines, and Kubernetes clusters.
- Designed a **Kubernetes-native deployment framework** supporting **50+ one-click applications** with tenant isolation, dynamic Helm value resolution, and Temporal-powered lifecycle management.

E2E Networks Ltd.

May 2025 – Nov 2025

Software Development Engineer Intern

New Delhi, India

- Automated deployment of production services, including GitLab, Mattermost and WordPress using Bash, eliminating manual provisioning.
- Developed backend APIs and business logic for the E2E Marketplace using Django REST Framework and integrating multiple third-party services.
- Automated the four-stage Marketplace product onboarding workflow, reducing onboarding time by **60%**.
- Configured Celery workers with RabbitMQ for scalable asynchronous task execution.

Inventive Core Integration Pvt. Ltd.

June 2021 – July 2021

IoT Data Analytics Intern

- Trained in industrial **automation and wearable technology**; developed a responsive website for **real-time data visualization** and comparative **graph plotting**.

Achievements

- **1st Position** – ESCI Collaboration with ICI (2023): “Mining real-time data & visualising COVID-19 probability based upon climatic conditions.”
- **GATE Qualified** – 2023

Education

Netaji Subhas University of Technology

2023–2025

M.Tech., Computer Science Engineering

CGPA: 7.68

Guru Jambheshwar University of Science and Technology, Hisar

CGPA: 7.51 2019–2023

B.Tech. in Electronics and Communication Engineering